

Coding Assignment 5

Due Date: *Monday, December 7 (11.59PM)*

Submission: on Gradescope

Instructions

The coding assignments may be completed in groups of *up to 4 students*. If you do so, please make sure that you include everyone's full name, and that you also *select everyone's name when submitting the assignment on Gradescope*. This ensures that each group member will get a grade assigned and have access to the comments from the graders.

For this assignment, the following items need to be submitted:

1. A **single PDF** that contains your code, all results, plots, and written answers (generated from R Markdown, Jupyter, etc.).
2. The **source file** used to produce the PDF (.Rmd, ipynb, ...).

Problem 1 [50 points]

In this problem we cluster countries using a 2022 snapshot of World Bank development indicators. The file `countries_2022.csv` is posted on Coursera. Each row is a country. The columns you may use as *features* are

`gdp_pc_ppp`, `fertility`, `life_exp`, `u5_mort`, `urban_pct`.

Do **not** use `region`, `income`, `population`, `iso3`, or `country` as features — those are labels for interpretation afterwards. Take a logarithm of `gdp_pc_ppp` before clustering (income is heavily right-skewed). Then scale (or standardize) all five features: the Euclidean metric in K -means is sensitive to units, and GDP would otherwise dominate fertility and mortality.

Our **goal** is to recover groups of countries with similar development profiles, and then to say where **Greece** sits in that partition.

- (a) Use **K -Means Clustering** to assign each country to a cluster:

- i. Use the **Gap Statistic** and the **Silhouette Statistic** to determine the **optimal number of clusters** k_{gap} and k_{sil} . Fit the K -means algorithm to the data for both k s. Use multiple random starts (e.g. `nstart/n_init` of at least 25).
Note: If you get the same k that is ok! You have more evidence that this is probably the appropriate number of clusters.
 - ii. Plot **fertility** vs. **life_exp** and color code the clusters for both k s. Mark **Greece** on the plot. Comment on the separability of the clusters. Keep in mind that this is a 2-dimensional projection of a 5-dimensional clustering.
 - iii. Interpret the clusters. Compute summary statistics for each cluster (mean fertility, life expectancy, under-5 mortality, urban share, and GDP per capita). Which cluster contains Greece? Name two other countries in that cluster and two countries that land far from Greece. Does the partition line up with the World Bank **region** or **income** labels, or does Greece sit with a mixed group?
- (b) Use **Hierarchical Clustering** to assign a cluster to each country. Use **complete linkage** and the same scaled features as in part (a).
- i. Plot the dendrogram. If the labels are unreadable, plot a truncated tree or label only a subset that includes Greece.
 - ii. Cut the dendrogram at k_{gap} and at k_{sil} from part (a) (if the two k s coincide, one cut is enough). Repeat questions (a-ii) and (a-iii) for these hierarchical assignments. Does Greece stay with the same neighbors as in part (a)? Comment on the differences and/or similarities.

Problem 2 [50 points]

In this problem we are going to create a simulated data set to illustrate the use of spectral clustering. In the literature, what we are going to simulate is known as the “Two Moons” data. Mathematically, each moon is a semicircle of radius r that can be parametrized in polar coordinates as

$$x = r \cos \theta, \quad y = r \sin \theta$$

for the upper half, and

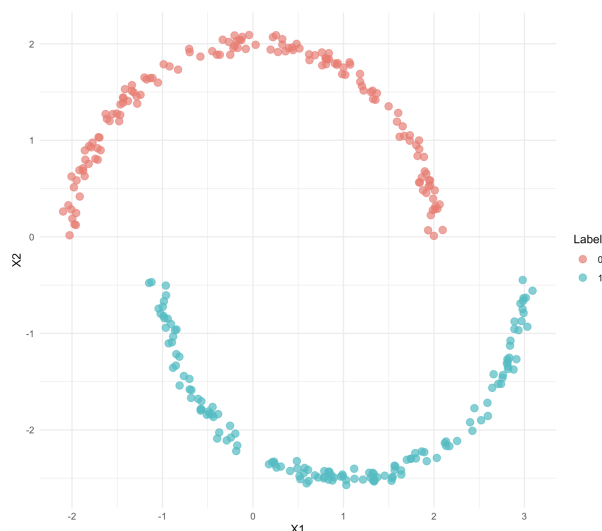
$$x' = dx - r \cos \theta, \quad y' = -r \sin \theta - dy$$

for the lower half, shifted by dx and dy so that the two semicircles do not overlap. Here $\theta \in [0, \pi]$ and r is the radius.

Create the *two moons* synthetic data of size $n = 200$ (100 points per moon) as follows:

1. For $r = 2$, sample $\theta_i^1 \sim \text{Uniform}[0, \pi]$, and compute $x_i = r \cos \theta_i^1$ and $y_i = r \sin \theta_i^1$. Add some noise to each coordinate, e.g. independent $\mathcal{N}(0, 0.05^2)$ draws (standard deviation 0.05, not variance 0.05). **Assign the label ‘0’ to this moon.**
2. For $r = 2$, sample $\theta_i^2 \sim \text{Uniform}[0, \pi]$, and compute $x'_i = 1 - 2 \cos \theta_i^2$ and $y'_i = -2 \sin \theta_i^2 - 0.5$. Add the same noise to each coordinate. **Assign the label ‘1’ to this moon.**
3. Combine everything into a data frame with 3 columns: the two coordinates and the generating label. This is your data set!
4. Plot it!

If you use $n = 200$ and `set.seed(598)` (in R), you should get something similar to the following:



Note: Python has a function that creates this synthetic data automatically, but you will lose all the fun! *It is ok to use it, but make sure that you plot your synthetic data.*

Now, let's apply clustering techniques to this data set. Use only the two coordinates as features — the generating label is for plotting and discussion, not an input to the algorithm.

- (a) Use ***K*-Means Clustering** to assign each point to a cluster (use $k = 2$). Plot the results by color-coding the data by cluster assignment. Comment on the results: does *K*-means recover the two moons? Why or why not? (Think about the geometry that *K*-means assumes.)
- (b) Use **Spectral Clustering** on the same data set (use $k = 2$). Plot the results by color-coding the data by cluster assignment. Comment on the results and compare with part (a) above.

Note: You are free to use the built-in R (`specc` in `kernlab`) or Python (`SpectralClustering` in `sklearn.cluster`) functions. In both cases you can use $k = 2$ and the Gaussian/RBF kernel.